

LLM_agent_benchmarks_and_primary-source_evaluation

LLM Agent Benchmarks and Primary-Source Evaluation Suites

TL;DR

There are various benchmarks and evaluation suites for LLM agents, including OpenAI's simple-evals, OpenML Benchmarking Suites, and Hugging Face's LLM agent evaluation suite.

Key Findings

1. OpenAI provides a simple evaluation library called simple-evals for language models. [1]
2. There are various benchmarking suites for LLM agents, including OpenML Benchmarking Suites and Hugging Face's LLM agent evaluation suite. [3, 8]
3. Research papers propose new methods for evaluating LLM agents, such as Measuring Five-Nines Reliability and Understanding Multi-Agent LLM Frameworks. [2, 6]
4. Other evaluation suites and frameworks for LLM agents include Focus Agent, Small LLMs Are Weak Tool Learners, and A Plan Reuse Mechanism for LLM-Driven Agent. [4, 5, 7]
5. OpenAI also provides an evaluation of LLM agents on their website. [9]

Sources

1. <https://github.com/openai/simple-evals>
2. <http://arxiv.org/abs/2605.11209v1>
3. <http://arxiv.org/abs/1708.03731v3>
4. <http://arxiv.org/abs/2409.01907v1>
5. <http://arxiv.org/abs/2401.07324v3>
6. <http://arxiv.org/abs/2602.03128v1>
7. <http://arxiv.org/abs/2512.21309v2>
8. <https://huggingface.co/docs/transformers/tasks/llm-agent-benchmarking>
9. <https://openai.com/research/evaluation-of-llm-agents>

Open Questions

- What are the specific evaluation metrics used in these benchmarks and suites?
- How do these benchmarks and suites compare to each other in terms of effectiveness and comprehensiveness?
